RealGas Simulator | Calculation record

Pure-component Peng-Robinson equation of state

Property (units in key)	Value
component	Methane
T_K	400
P_bar	50
T_ref_K	298.15
P_ref_bar	1
phase_rule	Stable
Z	0.9703643228
volume_m3_mol	0.000645444631
phi	0.967567967
fugacity_bar	48.37839835
H_ideal_J_mol	3900.819045
H_res_J_mol	-525.1751244
H_kJ_mol	3.37564392
S_ideal_J_molK	-21.31742928
S_res_J_molK	-1.038813251
S_J_molK	-22.35624253
thermal_error	None

Reference: H_ideal(Tref) = 0; S_ideal(Tref, Pref) = 0. Totals include departures at the calculated state and are not absolute properties or zeroed real-fluid differences.

Only one admissible root exists. A single root alone does not identify a liquid or vapour phase.

Data: Smith, Van Ness, Abbott & Swihart, 8e; App. B and Table C.1

See THEORY.md and the application calculation details for equations, derivative, units and derivation.